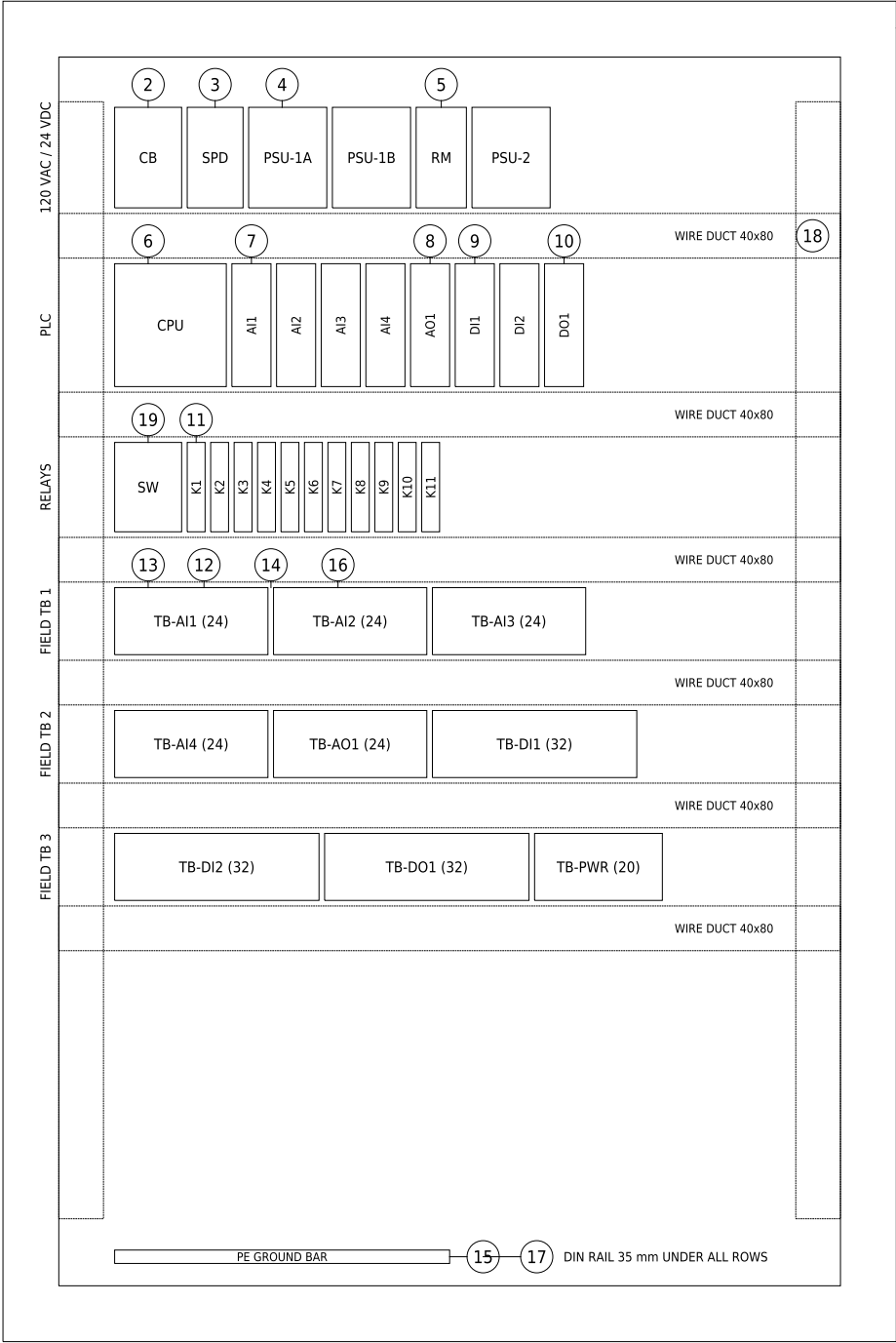


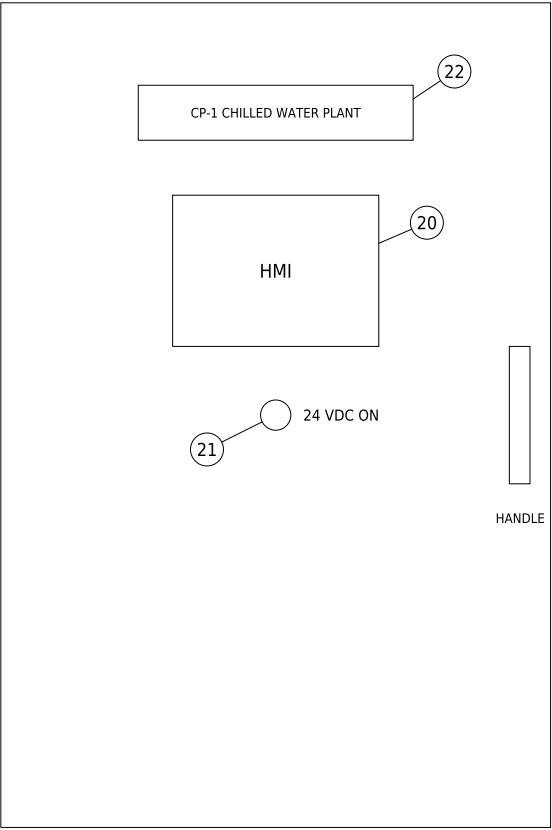
BACKPLATE VIEW SCALE 1:6
ENCLOSURE 1200 H x 800 W x 300 D, BACKPLATE 1100 x 700



BOTTOM GLAND PLATE: FIELD CABLE ENTRY

NOT FOR CONSTRUCTION - PORTFOLIO PROJECT

DOOR VIEW SCALE 1:10



NOTES

1. Built by a UL 508A listed panel shop; SCCR marked on nameplate.
2. Enclosure NEMA 12 (indoor, dust and drip tight).
3. Segregation: 120 VAC, 24 VDC analog and 24 VDC discrete wiring in separate ducts; cross at right angles only.
4. Spare: 20% minimum per I/O type, terminals installed for spare channels (88 channels for 64 points).
5. Field terminals grouped per I/O card (TB-AI1 ...); loop-powered AI fused per loop (26 fused terminals).
6. Shields grounded at panel end only (PE terminals).
7. 24 VDC: clean bus N+1 (PSU-1A/1B + RM), actuator bus PSU-2.
8. Heat: 37.6 C internal at 30 C ambient; no fan needed.
9. Fit check: every DIN row <= 80% full; see data/panel_bom.xlsx.

REV	DATE	DESCRIPTION
A	2026-10-08	ISSUED FOR REVIEW

BILL OF MATERIALS (full detail: data/panel_bom.xlsx)

ITEM	QTY	DESCRIPTION
1	1	Enclosure, NEMA 12, wall-mount, 1200 H x 800 W x 300 D mm, with backplate and...
2	1	Main circuit breaker, 120 VAC, 2-pole
3	1	Surge protective device, 120 VAC
4	3	Power supply 24 VDC, DIN-rail
5	1	Redundancy module 24 VDC (clean bus)
6	1	PLC CPU with Ethernet, BACnet/IP or Modbus TCP to BMS
7	4	Analog input module, 8-ch, 4-20 mA
8	1	Analog output module, 8-ch, 4-20 mA
9	2	Digital input module, 16-ch, 24 VDC
10	1	Digital output module, 16-ch, 24 VDC
11	11	Interposing relay, 24 VDC coil, 1 C/O contact, with base
12	26	Fused terminal block with 5 x 20 mm fuse, 0.5 A
13	170	Feed-through terminal block, 2.5 mm2
14	40	Ground (PE) terminal block, for cable shields
15	1	Ground bar, tinned copper
16	24	End plates, end stops and marker strips (sets)
17	3.6	DIN rail 35 mm (m)
18	6.4	Wire duct 40 x 80 mm with cover (m)
19	1	Ethernet switch, unmanaged, 5-port, DIN-rail
20	1	HMI touchscreen 10 in, door-mount
21	1	Pilot light, green, 24 VDC, "24 VDC ON"
22	1	Nameplates and wire markers (lot)

Data Center Chilled Water Plant - I&C Package

CONTROL PANEL CP-1 LAYOUT

DWG DC-E-101

REV A

DATE 2026-10-08 SCALE AS NOTED DRAWN S. Anandkumar

SHEET ANSI B